Anuj Yadav

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Education

University of Delhi — Delhi, India
Bachelor of Science (Hons.) in Electronics — *Expected May 2027*

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Database Management
- Operating Systems
- Digital Electronics
- Embedded Systems
- Communication Systems
- Microprocessors (8085)

Experience

Independent Embedded & Software Projects — Jan 2024 - Present
Self-directed engineering portfolio (25+ shipped applications) — *Delhi, India*

- Designed, built and deployed a public portfolio of 25+ working web applications and embedded systems, all hosted on GitHub Pages with every project live and verifiable by recruiters in one click.
- Developed a real-time 3D city engine from scratch in pure JavaScript - perspective projection, rotation matrices and painter's-algorithm depth sorting - rendering ~20,000 line segments and 2,000 solid faces per frame at 60fps with zero external libraries.
- Practiced production engineering discipline across the suite: per-app design systems, SEO metadata, WCAG-conscious accessibility, localStorage state persistence and XSS-safe input handling.

Projects

Smart Home Automation System | *Embedded C, Arduino, Proteus, GSM* — 2025

- Automated 4 home subsystems (fan, lighting, window, fire alarm) on a single ATmega328P by integrating LM35, PIR, MQ-3 and rain sensors with an L293D dual motor driver.
- Cut fire-alert response to under 3 seconds by driving a SIM900 GSM module over UART with raw AT commands for automatic SMS alerts alongside a local buzzer alarm.
- Republished the full Proteus circuit as an interactive web simulation that runs the actual firmware logic (LCD, pin states, motors) directly in the browser.

Checkmate - Chess Engine with AI | *JavaScript, Minimax, Alpha-Beta Pruning* — 2026

- Engineered a complete rules engine - legal move generation, castling, promotion, check/checkmate/stalemate - validated by 20/20 correct opening move generation.
- Implemented a minimax AI with alpha-beta pruning and piece-square evaluation across 3 difficulty levels; reduced search cost via capture-first move ordering.

Gridlock - Spreadsheet Formula Engine | *JavaScript, Recursive-Descent Parsing* — 2026

- Wrote a formula parser supporting cell references, ranges, SUM/AVG/MIN/MAX/IF and comparison operators with correct precedence, evaluated over a dependency graph with circular-reference detection.
- Shipped Excel-style UX: keyboard-first editing, formula bar, live aggregate statusbar, CSV import/export.

VoltLab - Digital Logic Simulator | *JavaScript, Canvas, Graph Simulation* — 2026

- Built a drag-and-wire circuit simulator (AND/OR/XOR/NAND gates, switches, LEDs) with live signal propagation, fixpoint iteration for feedback loops and automatic truth-table generation for up to 4 inputs.

AVR Embedded Practicals | *Embedded C, ATmega32, Bare-Metal* — 2025

- Implemented 15+ bare-metal programs covering hardware timers, interrupts, 4x4 keypad and HD44780 LCD interfacing, and buzzer control - simulated in Proteus and Wokwi.

Technical Skills

Languages: Python, C, C++, JavaScript (ES6+), SQL, HTML/CSS, 8085 Assembly
Embedded: Arduino UNO, ATmega32 (bare-metal AVR), UART/AT commands, L293D, sensor interfacing (LM35, PIR, MQ-3, rain), 16x2 LCD, timers & interrupts
Developer Tools: Git/GitHub, Proteus 8, Wokwi, MATLAB, Scilab, Google Colab, Linux (Termux), VS Code
Concepts: Data structures & algorithms, recursive-descent parsing, game-tree search, 3D graphics mathematics, DBMS, operating systems